MODEL QUESTION PAPER

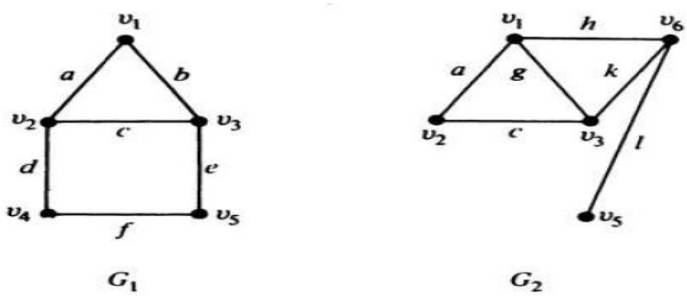
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
FOURTH SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR

Course Code: GAMAT401

Course Name: Mathematics for Computer and Information Science – 4

Max. Marks: 60		Duration: 2 hours 30 minutes

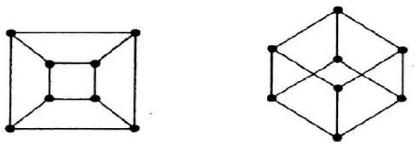
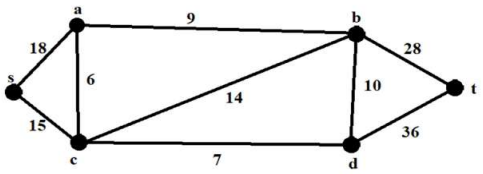
PART A

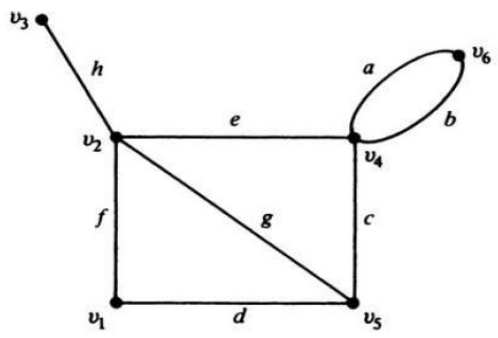
		Answer all questions. Each question carries (3) marks	CO	Marks
1		Prove that sum of the degrees of all vertices in G is twice the number of edges in G.	CO1	(3)
2		Define walk, path and circuit with examples.	CO1	(3)
3		Find the union of $G_1 and\ G_2$ and ring sum of $G_1\ and\ G_2$ 	CO2	(3)
4		Define Euler graph. Give an example of Eulerian graph which is not Hamiltonian.	CO2	(3)
5		Draw all unlabelled trees with 5 vertices.	CO3	(3)
6		Find the maximum number of vertices possible in a 3-level binary tree. Also find the maximum height possible in a binary tree with 11 vertices.	CO3	(3)
7		Show that every cut set in a connected graph must contain at least one branch of every spanning tree of G.	CO4	(3)
8		Prove that edge connectivity of a graph cannot exceed the degree of the vertex with the smallest degree in G.	CO4	(3)

PART B

Answer any one full question from each module. Each question carries (9) marks

Module 1

9	a)	 Are the two graphs isomorphic ? Justify your answer.	CO1	(3)
	b)	Show that a simple graph with n vertices must be connected if it has more than $\frac{(n-1)(n-2)}{2}$ edges.	CO1	(6)
10	a)	Define complete graph and complete bipartite graph. Draw a graph which is a complete graph as well as a complete bipartite graph.	CO1	(4)
	b)	If a graph has exactly two vertices of odd degree, then prove that there must be a path joining these two vertices.	CO1	(5)
		Module 2		
11	a)	Define directed graphs. Differentiate between symmetric digraphs and asymmetric digraphs	CO2	(3)
	b)	Nine members of a new club meet each day for lunch at a round table. They decide to sit such that every member has different neighbours at each lunch. How many days can this arrangement last? Justify your answer.	CO2	(6)
12	a)	Define Hamiltonian circuit and Hamiltonian path. Give an example for each.	CO2	(3)
	b)	Prove that a connected graph is Euler if and only if it can be decomposed into circuits.	CO2	(6)
		Module 3		
13		Write down Dijkstra's algorithm and use it to find the shortest path from s to t.  	CO3	(9)
14		Show that a Tree with n vertices has exactly n-1 edges.	CO3	(9)
		Module 4		
15		Define cut-set. Prove that every circuit in G has an even number of edges in common with any cut-set.	CO4	(9)

16		For the following graph, find the (i) Incidence matrix (ii) Adjacency matrix (iii) Path matrix between $v_1 and\ v_4$ (iv) Circuit matrix  	CO4	(9)
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MODEL QUESTION PAPER

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
FOURTH SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR

Course Code: PCITT402

Course Name: COMPUTER NETWORKS

Max. Marks: 60			Duration: 2 hours 30 minutes

		PART A		
		Answer all questions. Each question carries 3 marks	CO	Marks
1		Describe how Computer Networks are classified	1	(3)
2		Differentiate between switch, router and gateway.	1	(3)
3		Describe the framing method 'Flag bytes with byte stuffing'	2	(3)
4		Describe the concept of sliding window protocol.	2	(3)
5		State the optimality principle and describe its significance in the routing process.	3	(3)
6		Differentiate between Class A, Class B, Class C IPv4 addresses.	3	(3)
7		List and describe the primitives defined for a simple transport service	4	(3)
8		Draw the UDP segment structure and describe the fields.	4	(3)
		PART B		
		Answer any one full question from each module. Each question carries 9 marks		
		Module 1		
9	a)	Draw the OSI reference model showing its layers and its various elements. Explain the functions of each layer.	1	(9)
10	a)	Explain the different guided transmission media used for transmission of data over computer networks.	1	(9)
		Module 2		
11	a)	Give the algorithm for the error detection mechanism **Cyclic Redundancy Check.** In a data communication system that uses CRC for error detection; a sender wishes to send a frame 110101111 with a generator polynomial $x^4 + x + 1$. Calculate the frame transmitted after performing the CRC calculation.	2	(6)
	b)	Describe the Collision Free Protocol token ring.	2	(3)

12	a)	Draw the Ethernet frame format and describe the fields	2	(5)
	b)	Compare and contrast 1-persistent CSMA, p-persistent CSMA, non-persistent CSMA, and CSMA with collision detection (CSMA/CD).	2	(4)
Module 3				
13	a)	Draw the IPv4 header format and explain each of its fields in detail, highlighting their significance and purpose.	3	(7)
	b)	Explain the functioning of choke packets and how they help in reducing network congestion.	3	(2)
14	a)	Describe the Distance Vector routing algorithm and demonstrate how a router modifies its routing table using an appropriate example.	3	(7)
	b)	Explain the purpose and operation of the Hop Limit field in the IPv6 header.	3	(2)
Module 4				
15	a)	Describe the Domain Name System (DNS) and explain how it functions to resolve human-readable domain names into machine-readable IP addresses. Include its hierarchical structure of the name space.	4	(6)
	b)	List and describe the commonly used built-in HTTP request methods	4	(3)
16	a)	Draw a labeled diagram of the TCP segment header and briefly describe the function of each field	4	(7)
	b)	Describe the purpose of Sequence number and Synchronization source identifier fields present in the RTP protocol header.	4	(2)
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MODEL QUESTION PAPER

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
FOURTH SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR

Course Code: PCITT403

Course Name: OPERATING SYSTEM

Max. Marks: 60		Duration: 2 hours 30 minutes

PART A

		Answer all questions. Each question carries 3 marks	CO	Marks
1		What are the three methods for passing parameters to the OS in System Calls?	1	(3)
2		The long-term scheduler controls the degree of programming. Justify your answer.	1	(3)
3		What are the different scheduling criteria?	2	(3)
4		What are the necessary conditions for a deadlock situation?	2	(3)
5		What is demand paging? What are its advantages?	3	(3)
6		Explain Belady's anomaly with an example.	3	(3)
7		Differentiate between single level structure and two-level directory structure.	4	(3)
8		Define seek time, rotational latency and transfer time in disk scheduling.	4	(3)

PART B

Answer any one full question from each module. Each question carries 9 marks

Module 1

9	a)	Explain the different types of Operating Systems.	1	(9)
10	a)	Consider the following set of 5 processes, whose arrival time, burst time and priority are given:	1	(9)

Process	P1	P2	P3	P4	P5
Arrival Time	0	1	2	3	4
Burst Time	4	3	1	5	2
Priority	2	3	4	5	5

Draw Gantt Chart for the execution of these processes and find the average waiting time and average turnaround time (Higher number a represents higher priority), if the CPU scheduling policy is (i) priority preemptive (ii) SRTF

Module 2				
11	a)	What is semaphore? How producer consumer problem is solved using semaphore?	2	(9)
12	a)	Consider the following snapshot of a system with five processes P1, P2, P3, P4, P5 and four resources A, B, C, D	2	(9)

Process	Max	Allocation	Available
	A B C D	A B C D	A B C D
P1	0 0 1 2	0 0 2 3	1 3 2 2
P2	1 0 0 1	1 6 4 2	
P3	1 2 4 2	2 3 6 5	
P4	0 5 1 1	0 6 2 5	
P5	0 0 1 2	1 6 5 3	

Use Banker's algorithm to answer:

i) How many instances of A, B, C, D are there?

ii) What is the content of need matrix?

iii) Is the system in a safe state? If it is, find the safe sequence.

Module 3				
13	a)	Illustrate paging and segmentation with a suitable diagram.	3	(9)
14	a)	What is Compaction?	3	(3)
	b)	Consider the page references 7,0,1,2,0,3,0,4,2,3,0,3,2. Using demand paging with three frames, how many page faults would occur for the following page replacement algorithms: i) LRU replacement ii) FIFO replacement iii) Optimal	3	(6)
Module 4				
15	a)	Consider a disk drive has 200 cylinders numbered from 0 to 199 and the current position of the head is at cylinder number 65. The requests for I/O to blocks are on cylinders 55, 68, 183, 39, 18, 190, 160, 150, 98. Draw the head movement	4	(9)

		in FCFS, SSTF, SCAN disk scheduling algorithms and compute the total head movements (in cylinders) in each.		
	b)			
16	a)	Describe various file access methods.	4	(5)
	b)	How are directories implemented?	4	(4)
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MODEL QUESTION PAPER

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
FOURTH SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR

Course Code: PBITT404

Course Name: DATA SCIENCE

Max. Marks: 40		Duration: 2 hours 30 minutes

PART A

		Answer all questions. Each question carries 2 marks	CO	Marks
1		Define Data Science. How is it different from traditional data analysis?	CO1	(2)
2		Name any three commonly used Python packages in data science and state one purpose of each.	CO1	(2)
3		Mention the purpose of a data pipeline in a data science workflow?	CO2	(2)
4		What is dimensionality reduction? Mention one technique used for it.	CO2	(2)
5		What is the key idea behind rule-based classification using IF-THEN rules?	CO3	(2)
6		Name any two distance measures used in clustering algorithms.	CO3	(2)
7		Mention any two metrics used to evaluate a classification model.	CO4	(2)
8		What does the ROC curve represent?	CO4	(2)

PART B

Answer any one full question from each module. Each question carries 6 marks

Module 1

9	a)	What are the steps involved in installing the latest version of Python on a Windows system?	CO1	(2)
	b)	List and explain any four commonly used Python packages for data science. Mention one key feature of each.	CO1	(4)
10	a)	Explain any two basic data pre processing techniques using pandas.	CO1	(3)
	b)	Describe the process of loading a dataset using the pandas library in Python. Write a sample code snippet to load a CSV file.	CO1	(3)

Module 2

11	a)	Define hyperparameter optimization. Name one common method used for it.	CO2	(2)
	b)	What is Exploratory Data Analysis (EDA)? Explain the role of scatter plots in understanding relationships in data with a suitable example.	CO2	(4)
12	a)	Why is feature selection important in machine learning models?	CO2	(2)
	b)	What is cross-validation in the context of model validation? How does it help in improving model performance?	CO2	(4)

Module 3				
13	a)	Explain the working of the K-Nearest Neighbour (K-NN) algorithm with a suitable example. What role does the distance measure play in K-NN?	CO3	(4)
	b)	State one key difference between supervised and unsupervised learning.	CO3	(2)
14	a)	Differentiate between Agglomerative and Divisive Hierarchical Clustering. Illustrate with a diagram or example.	CO3	(4)
	b)	Mention one difference between linear regression and logistic regression.	CO3	(2)
Module 4				
15	a)	A binary classification model was tested on 100 samples and produced the following confusion matrix: 40 true positives, 10 false negatives, 20 false positives, and 30 true negatives. Based on this, calculate the model's precision**,** recall, and F1-score	CO4	(4)
	b)	Define sensitivity and specificity in the context of model evaluation.	CO4	(2)
16	a)	Describe the steps involved in building and evaluating your first data science model. How do you interpret the model's performance?	CO4	(4)
	b)	Differentiate between True Positive (TP) and False Positive (FP) in the context of a confusion matrix.	CO4	(2)
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MODEL QUESTION PAPER

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
FOURTH SEMESTER B. TECH DEGREE EXAMINATION, MAY 2026

Course Code:PEITT 411

Course Name: Object Oriented Design using JAVA

Max. Marks: 60		Duration: 2 hours 30 minutes

PART A

		Answer all questions. Each question carries 3 marks	CO	Marks
1		Differentiate Functional and Non-functional Requirements in Software Engineering Concepts.	CO1	(3)
2		What is UML?	CO1	(3)
3		Explain on Requirements elicitation	CO2	(3)
4		Write a short note on Analysis Model.	CO2	(3)
5		Define Coupling and Cohesion.	CO3	(3)
6		Explain Delegation.	CO3	(3)
7		Explain Class Implementor, Class Extender, and Class User.	CO4	(3)
8		Explain Forward engineering and Reverse engineering.	CO4	(3)

PART B

Answer any one full question from each module. Each question carries 9 marks

Module 1

9	a)	Explain Notations, Methods, and Methodologies in Software Engineering Concepts.	CO1	(6)
	b)	Write a short note on Communication in Managing Software Development.	CO1	(3)
10	a)	Explain Class diagram with suitable example.	CO1	(5)
	b)	Explain UML State machine diagram.	CO1	(4)

Module 2

11	a)	Explain Identifying Actors.	CO2	(5)
	b)	Explain Identifying Relationships among Actors and Use Cases in Requirements Elicitation Activities.	CO2	(4)
12	a)	Explain Entity, Boundary, and Control Objects with suitable example.	CO2	(6)
	b)	Differentiate Generalization and Specialization with suitable example.	CO2	(3)

Module 3

13	a)	Explain Layers and Partitions.	CO3	(3)
	b)	Explain Model/View/Controller (MVC) architectural style.	CO3	(6)
14	a)	Explain Application frameworks.	CO3	(6)
	b)	Explain Documenting Reuse.	CO3	(3)
Module 4				
15	a)	List out the Interface specification activities of object design.	CO4	(5)
	b)	Write a short note on Object Constraint Language (OCL).	CO4	(4)
16	a)	Explain Mapping Associations to Collections.	CO4	(5)
	b)	List out the Transformation Principles.	CO4	(4)
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MODEL QUESTION PAPER

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
FOURTH SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR

Course Code: PEITT414

Course Name: DATA COMMUNICATION AND NETWORKING

Max. Marks: 60		Duration: 2 hours 30 minutes

PART A

		Answer all questions. Each question carries 3 marks	CO	Marks
1		Write a short note on broadcast radio	1	(3)
2		Define Nyquist theorem. Consider a noiseless channel with a bandwidth of 3000 Hz transmitting a signal with two signal levels.What is the maximum bit rate?	1	(3)
3		Differentiate between even parity and odd parity with suitable examples.	2	(3)
4		Discuss various steps involved in Pulse Code Modulation	2	(3)
5		Differentiate between Frequency Division multiplexing and Time division multiplexing	3	(3)
6		Draw the general model of the Spread Spectrum digital communication system. Give a brief description of the components.	3	(3)
7		Describe the elements of a Circuit-Switch Node	4	(3)
8		Prepare a short note on Flooding.	4	(3)

PART B

Answer any one full question from each module. Each question carries 9 marks

Module 1

9	a)	With a neat diagram explain the data communication model	1	(6)
	b)	Describe Data rate, Noise and Bandwidth with respect to a channel.	1	(3)
10	a)	Explain various guided transmission medias for data communication	1	(9)

Module 2

11	a)	In a data communication system that uses CRC for error detection; a sender wishes to send a frame 110101111 with a generator polynomial x^4+x+1. Calculate the frame transmitted after performing the CRC calculation.	2	(5)
	b)	Draw the following line coding schemes for representing the data 01001110 (1) NRZ-I (2) NRZ-L	2	(4)

12	a)	With neat diagramsDescribe various techniques for converting digital data to analog signals	2	(6)
	b)	Draw the encoding scheme for the bit stream 1100001000000000 using HDB3 technique.	2	(3)
Module 3				
13	a)	Demonstrate the working of the sliding window method for flow control with suitable examples.	3	(5)
	b)	Draw the block diagram and describe the transmitter side of the Frequency Hopping Spread Spectrum.	3	(4)
14	a)	Explain synchronous Time Division multiplexing system with a neat diagram.	3	(5)
	b)	Describe the Go-back-N ARQ scheme of error control with suitable example.	3	(4)
Module 4				
15	a)	With neat diagrams explain the different types of Packet switching architectures.	4	(6)
	b)	Describe the parameters commonly used in characterizing frequency reuse in cellular networks.	4	(3)
16	a)	Compare softswitch architecture with Traditional circuit switching architecture. Draw neat diagrams for both.	4	(6)
	b)	Describe the adaptive routing strategy	4	(3)
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MODEL QUESTION PAPER

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
FOURTH SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR

Course Code:PEITT413

Course Name: FOUNDATIONS OF SECURITY

Max. Marks: 60		Duration: 2 hours 30 minutes

PART A

		Answer all questions. Each question carries 3 marks	CO	Marks
1		Define the greatest common divisor of two integers. Which algorithm can effectively find the GCD?	CO1	(3)
2		Explain the algebraic structure field.	CO1	(3)
3		Define Fermat's little theorem and explain its application	CO2	(3)
4		What are Mersenne primes?	CO2	(3)
5		Define the Jacobi symbol.	CO3	(3)
6		Find the result of 12^{-1} □□□ 77using Eulers theorem	CO3	(3)
7		Explain Caesar cipher with an example.	CO4	(3)
8		List the difference between stream cipher and block cipher.	CO4	(3)

PART B

Answer any one full question from each module. Each question carries 9 marks

Module 1

9	a)	Find the greatest common divisor of 2740 and 1760	CO1	(3)
	b)	Write Extended Euclidean algorithm	CO1	(3)
	c)	Find the multiplicative inverse of 23 in Z 100.	CO1	(3)
10	a)	Explain the algebraic structures groups and ring.	CO1	(4)
	b)	What is Lagrange's theorem?	CO1	(3)
	c)	What is order of a group?	CO1	(2)

Module 2

11	a)	Show that whether the number 561,100,109 pass the Miller-Rabin test?	CO2	(4)
	b)	Find the value of x for the following sets of congruence using the Chinese remainder theorem. x ≡ 2 mod 7 and x≡ 3 mod 9	CO2	(5)

12	a)	How many of the following integers pass the Fermat primality test:100,110,130,150,200	CO2	(5)
	b)	Determine how many integers pass Miller Rabin primality test: 200,250,271	CO2	(4)
Module 3				
13	a)	Define Euler totient function and its application	CO3	(3)
	b)	Define quadratic congruence and the importance of QRs and QNRs in solving quadratic equations.	CO3	(3)
	c)	Using quadratic residue solve the following congruence: $\square^{2} \equiv 4 \square\square\square 7$	CO3	(3)
14	a)	Using quadratic residue solve the following congruences: $\square^{2} \equiv 4 \square\square\square 14$ $\square^{2} \equiv 5 \square\square\square 10$	CO3	(5)
	b)	Find the results of the following using euler's theorem $16^{-1} \square\square\square 323$,$20^{-1} \square\square\square 403$	CO3	(4)
Module 4				
15	a)	Encrypt the text "welcome to network security" using a transposition cipher with the key 'LAYER'	CO4	(5)
	b)	Distinguish between diffusion and confusion.	CO4	(2)
	c)	What is one time pad?	CO4	(2)
16	a)	Use an Affine cipher to encrypt the message 'its cool' with the key pair(5,8) in modulus 26.	CO4	(3)
	b)	Distinguish between substitution and transposition ciphers	CO4	(3)
	c)	What is hill cipher and playfair cipher?	CO4	(3)
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MODEL QUESTION PAPER

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
FOURTH SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR

Course Code: PEITT414

Course Name: COMPUTER GRAPHICS

Max. Marks: 60		Duration: 2 hours 30 minutes

PART A

		Answer all questions. Each question carries 2 marks	CO	Marks
1		What is the refresh rate in raster scan systems? Explain its significance.	CO1	(3)
2		Write the condition to choose the next pixel in Bresenham's algorithm.Explain its logic	CO1	(3)
3		What is the purpose of using homogeneous coordinates in 2D transformations?	CO2	(3)
4		What is the purpose of the window-to-viewport transformation?	CO2	(3)
5		Distinguish between exploded and cutaway views.	CO3	(3)
6		What is the function of the viewing transformation in 3D graphics?	CO3	(3)
7		Mention any two advantages of the A-buffer method over the Z-buffer.	CO4	(3)
8		What is morphing in computer graphics?Explain with its applications.	CO4	(3)

PART B

Answer any one full question from each module. Each question carries 9 marks

Module 1

9	a)	Compare OLED, AMOLED, and E-Paper displays in terms of structure, efficiency, and applications.	CO1	(4)
	b)	Explain the Digital Differential Analyzer (DDA) line drawing algorithm with a suitable example.	CO1	(5)
10	a)	Explain the midpoint circle generation algorithm. Derive the decision parameters used for pixel selection.	CO1	(4)
	b)	Describe flood fill and boundary fill algorithms. Compare them with respect to input, performance, and limitations.	CO1	(5)

Module 2

11	a)	Show calculations with transformation matrices for a triangle has vertices A(1, 1), B(3, 1), C(2, 4). Apply the following transformations: i) Reflect the triangle about the X-axis. ii) Perform a shear in the X-direction with shear factor = 1.	CO2	(4)
	b)	**Clip a line segment with endpoints A(2, 3) and B(10, 7) using the Cohen-Sutherland line clipping algorithm for a window defined by xmin = 4, xmax = 8, ymin=2,ymax=6.***(Show region codes, decision steps, and clipped output.)*	CO2	(5)
12	a)	Using the Sutherland-Hodgeman polygon clipping algorithm, clip a polygon defined by vertices (1, 1), (5, 1), (5, 5), (1, 5) against a rectangular window with boundaries xmin = 2, xmax = 4, ymin = 2, ymax = 4.	CO2	(4)
	b)	Explain the Weiler-Atherton polygon clipping algorithm. Given a concave polygon and a rectangular clipping window, show how the algorithm handles entry and exit points with a diagram.	CO2	(5)
Module 3				
13	a)	Explain perspective and parallel projections with diagrams. Show how a 3D point (x, y, z) is projected onto a 2D plane in each case.	CO3	(4)
	b)	Describe the process of visible surface detection using the Z-buffer algorithm. Apply it conceptually to a scene with two overlapping rectangles.	CO3	(5)
14	a)	Explain depth cueing and surface rendering. How do they improve realism in 3D graphics? Illustrate with a simple example.	CO3	(4)
	b)	Describe the steps in viewing transformation in 3D graphics. Show how a 3D scene is converted from world coordinates to view coordinates.	CO3	(5)
Module 4				
15	a)	Explain the Z-buffer algorithm. Describe how it works with an example of overlapping polygons.	CO4	(4)
	b)	Explain the area-subdivision method for hidden surface elimination with a recursive example.	CO4	(5)
16	a)	Explain the basic principles of animation. Describe keyframe animation and tweening with diagrams.	CO4	(4)
	b)	Define digital image processing. Explain the basic steps involved in DIP with a neat block diagram.	CO4	(5)

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MODEL QUESTION PAPER

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
FOURTH SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR

Course Code:PEITT 415

Course Name: OPERATION RESEARCH

Max. Marks: 60		Duration: 2 hours 30 minutes

PART A

		Answer all questions. Each question carries 3 marks	CO	Marks
1		A company sells two different products A and B, making a profit of Rs 40 and Rs 30 per unit, respectively. They are both produced with the help of a common production process and are sold in two different markets. The production process has a total capacity of 30,000 man-hours. It takes three hours to produce a unit of A and one hour to produce a unit of B. The market has been surveyed and company officials feel that the maximum number of units of A that can be sold is 8,000 units and that of B is 12,000 units. Subject to these limitations, products can be sold in any combination. Formulate this problem as an LP model to maximize profit.	CO1	(3)
2		Use the graphical method to solve the following LP problem. Minimize $\square = 3\square_1 + 2\square_2$ subject to the constraints (i) $5\square_1 + \square_2 \geq 10$, (ii) $\square_1 + \square_2 \geq 6$, (iii) $\square_1 + 4\square_2 \geq 12$ and $\square_1, \square_2 \geq 0$.	CO1	(3)
3		What do you understand by the term 'sensitivity analysis'? Discuss the effect of (i) variation of cj , (ii) variation of bi and (iii) addition of a new constraint.	CO2	(3)
4		What is the principle of duality in linear programming? Explain its advantages.	CO2	(3)
5		With reference to a transportation problem define the following terms: (i)Feasible solution (ii) Basic feasible solution (iii) Optimal solution (iv) Nondegenerate basic feasible solution	CO3	(3)
6		Give the mathematical formulation of an assignment problem. How does it differ from a transportation problem?	CO3	(3)
7		Discuss the difference between decision-making under certainty, under uncertainty and under risk.	CO4	(3)
8		Consider the game with the following payoff table:	CO4	(3)

		Player B Player A | B_1 B_2 A_1 | 2 6 A_2 | -2 λ Show that the game is strictly determinable, whatever λ may be. And, determine the value of the game.		
		PART B		
		Answer any one full question from each module. Each question carries 9 arks m		
		Module 1		
9		A company manufactures two kinds of machines, each requiring a different manufacturing technique. The deluxe machine requires 18 hours of labour, 8 hours of testing and yields a profit of Rs 400. The standard machine requires 3 hours of labour, 4 hours of testing and yields a profit of Rs 200. There are 800 hours of labour and 600 hours of testing available each month. A marketing forecast has shown that the monthly demand for the standard machine is to be more than 150. The management wants to know the number of each model to be produced monthly that would maximize total profit. Formulate and solve this as a linear programming problem.	CO1	(9)
10		Consider the following problem: Maximize $\square = 3\square_1 + 2\square_2 + 3\square_3$ subject to $2\square_1 + \square_2 + \square_3 = 4$ $\square_1 + 3\square_2 + \square_3 = 12$ $3\square_1 + 4\square_2 + 2\square_3 = 16$ $\square_1, \square_2, \square_3 \geq 0$	CO1	(9)
	a)	Show that Phase I will terminate with an artificial basic variable at zero level	CO1	
	b)	Remove the zero artificial variable prior to the start of Phase II, then carry out Phase II iterations.	CO1	
		Module 2		

11		A company produces two products, A and B. The unit revenues are \$2 and \$3, respectively. Two raw materials, M1 and M2, used in the manufacture of the two products have daily availability of 8 and 18 units, respectively. One unit	CO2	(9)
		of A uses 2 units of M1 and 2 units of M2, and 1 unit of B uses 3 units of M1 and 6 units of M2.		
	a)	Determine the dual prices of M1 and M2 and their feasibility ranges.	CO2	
	b)	If M2 availability is increased by 3 units, determine the associated optimum revenue		
	c)	Suppose that 2 additional units of M1 can be acquired at the cost of 25 cents per unit. Would you recommend the additional purchase and why?	CO2	
12	a)	A company makes three products: X, Y and Z out of three raw materials A, B and C. The raw material requirements are given below:	CO2	(9)

Raw Materials	Number of Units of Raw Material Required to Produce One Unit of Product		
	X	Y	Z
A	1	2	1
B	2	1	4
C	2	5	1

The unit profit contribution of the products: X, Y and Z is Rs. 40, 25 and 50, respectively. The number of units of raw material available are 36, 60 and 45, respectively.

(a) Formulate this problem as a linear programming problem.

(b) Write its dual.

(c) Solve the dual by the simplex method and find the optimal product mix and the shadow prices of the raw material and labour.

Module 3

13	a)	Prove that the necessary and sufficient condition for the existence of a feasible solution to the transportation problem is	CO3	(9)

$\sum_{i=1}^{m} a_i = \sum_{j=1}^{n} b_j$ (Rim condition)

That is, the total capacity (or supply) must equal total requirement (or demand).

14	a)	Five men are available to do five different jobs. From past records, the time (in hours) that each man takes to do each job is known and is given in the following table:	CO3	(9)

		Jobs				
		I	*II*	*III*	*IV*	*V*
Men	*A*	2	9	2	7	1
	B	6	8	7	6	1
	C	4	6	5	3	1
	D	4	2	7	3	1
	E	5	3	9	5	1

Find out how men should be assigned the jobs in a way that will minimize the total time taken.

Module 4

15	a)	Mr Sethi has Rs 10,000 to invest in one of three options: A, B or C. The return on his investment depends on whether the economy experiences inflation, recession, or no change at all. The possible returns under each economic condition are given below:	CO4	(4)

	State of Nature		
Strategy	*Inflation*	*Recession*	*No Change*
A	2,000	1,200	1,500
B	3,000	800	1,000
C	2,500	1,000	1,800

What should he decide, using the pessimistic criterion, optimistic criterion, equally likely criterion and regret criterion?

	b)	XYZ Company manufactures guaranteed tennis balls. At present, approximately 10 per cent of the tennis balls are defective. A defective ball leaving the factory costs the company Re 0.50 to honour its guarantee. Assume that all defective balls are returned. At a cost of Re 0.10 per ball, the company can conduct a test that always correctly identifies both good and bad tennis balls. (a) Draw a decision tree and determine the optimal course of action and its expected cost. (b) At what test cost should the company be indifferent to testing?	CO4	(5)

16	a)	Assume that two firms are competing for the market share for a particular product. Each firm is considering what promotional strategy to employ for the coming period. Assume that the following payoff matrix describes the increase in market share of Firm A and the decrease in market share for Firm B. Determine the optimal strategies for each firm.	CO4	(9)

	Firm B		
Firm A	*No Promotion Promotion*	*Moderate Promotion*	*Much*
No Promotion	5	0	–10
Moderate Promotion	10	6	2
Much Promotion	20	15	10

a. Which firm would be the winner, in terms of market share?

b. Would the solution strategies necessarily maximize the profits for either of the firms?

c. What might the two firms do to maximize their profits?

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MODEL QUESTION PAPER

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

FOURTH SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR

Course Code: UCHUT346

Course Name: Economics for Engineers

Max. Marks: 50		Duration: 2 hours 30 minutes

PART A

		Answer all questions. Each question carries 3 marks	CO	Marks
1		What are the central problems of an economy?	CO1	(3)
2		Point out any three applications of price elasticity of demand.	CO1	(3)
3		What is the social cost of production?	CO2	(3)
4		What is repo rate?	CO3	(3)
5		What is esteem value?	CO4	(3)
6		Write a short note on time value of money.	CO4	(3)

PART B

Answer any one full question from each module. Each question carries 8 marks

Module 1

9	a)	Suppose a country is producing at a point inside the production possibility curve. Draw a PPC and examine this situation.	CO1	(5)
	b)	State the law of demand. Point out any two exceptions of this law.	CO1	(3)

10	a)	A consumer purchases 10 units of a commodity when its price is Rs.100. Later when its price falls to Rs.90, he purchases 8 units only. Estimate price elasticity. What type of a commodity is this?	CO1	(5)
	b)	State the law of variable proportion.	CO1	(3)

Module 2				
11	a)	What is oligopoly? Why price is rigid under oligopoly?	CO2	(5)
	b)	The cost function of a firm is given as $TC=1000+10Q-6Q^2+Q^3$.. Calculate fixed cost, variable cost and marginal cost when output is 10 units.	CO2	(3)
12	a)	Suppose a firm is earning super normal profit under monopolistic market condition. Explain this situation by drawing a diagram.	CO2	(5)
	b)	Suppose a firm sells its product at a price of Rs.10 per unit and its average variable cost is Rs.6. If the firm spend Ra.10000 as rent and pay Rs. 6000 as interest every month, estimate its break-even level of output.	CO2	(3)
Module 3				
13	a)	What is inflation? How does inflation affect investment and production.	CO3	(5)
	b)	How will you obtain NNPfc from GDPmp.	CO3	(3)
14	a)	From the data given below (In Rs. Crores) estimate GDPmp and national income. Private final consumption expenditure = 1000, Government expenditure = 500, Invest expenditure = 700, Net exports = 300, Depreciation = 200, NFIA=(-200) and Net indirect tax = 100	CO3	(5)

	b)	What is bank rate? Examine the bank rate policy of the government during inflation.	CO3	(3)
		Module 4		
15	a)	Examine the procedures of value engineering.	CO4	(5)
	b)	Examine the application areas of value engineering	CO4	(3)
16	a)	1. Suppose the initial investment of a project is Rs. 3000 (Crores) and the cost of capital or the opportunity cost of capital is 10 percent. Calculate NPV of the project based on the cash flows given below. Year 1 2 3 4 5 Cash flow 1000 900 800 700 600 (In Crores)	CO4	(5)
	b)	Point out any three merits of NPV method.	CO4	(3)

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MODEL QUESTION PAPER

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR

Course Code: UCHUT347

Course Name: ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT

Max. Marks: 50		Duration: 2 hours 30 minutes

PART A

		Answer all questions. Each question carries 3 marks	CO	Marks
1		What is civic virtue? How does it relate to professional ethics?	1	(3)
2		What do you mean by 'gendered technology' and provide one specific example of a product whose design or application reflects this concept.	2	(3)
3		Explain the term triple bottom line in sustainable engineering.	3	(3)
4		Why is biodiversity considered essential for ecosystem stability? Give one example.	3	(3)
5		What is the idea behind degrowth? How is it different from economic development?	4	(3)
6		Identify two engineering solutions that can reduce greenhouse gas emissions in everyday life.	5	(3)

PART B

Answer any one full question from each module. Each question carries 8 marks

Module 1

9		You are part of a team designing an app that collects user data for public services. What ethical concerns would you raise regarding data privacy and consent? Suggest safeguards.	1	(8)
10	a)	What is the fundamental difference between 'Equity' and 'Equality' when discussing gender initiatives and resource allocation in education/workplace?	2	(5)
	b)	Suggest measures to promote gender equity in workplaces.	2	(3)

Module 2

11		A new industrial project near a wetland promises jobs but threatens local ecosystems. Apply the three ethical lenses (anthropocentric, biocentric, ecocentric) to assess it and recommend a balanced decision.	3	(8)
12	a)	Explain how life-cycle analysis (LCA) can help engineers design more sustainable products.	3	(5)
	b)	Illustrate with one example relevant to Kerala.	3	(3)

Module 3				
13		Rapid motorization is worsening air pollution. Suggest sustainable mobility options suitable for medium-sized Kerala towns, explaining their environmental benefits.	4	(8)
14		Outline a three-step plan to move a college campus towards zero waste. Mention the role of engineers, students, and the college management in each step.	4	(8)
Module 4				
15	a)	Discuss how renewable energy technologies (solar, wind, biogas) can be scaled up in Kerala.	5	(4)
	b)	State two technical and two social challenges to their adoption.	5	(4)
16	a)	Discuss the importance of environmental policies and regulations in promoting sustainable engineering.	5	(4)
	b)	Give one national and one international example and their impact.	5	(4)
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